Dictionary study: does a larger, in-network lift help?

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Summary. Does a larger dictionary with in-network measurements, as in Max Sibeijn's paper on this AROMA benchmark, predict or control better than the production 47-feature lift? No. At the horizons the controller uses, and on the trajectory it actually visits, the broad in-network lift predicts worse, by about +220 to +260 W delivered-heat RMSE, whichever of the two controllers generated that trajectory. A leaner targeted subset lowers the suite average but only helps the large consumers, not the worst-served one that sets comfort, so it does not change the choice. The one-step and temperature-metric wins are conceded below. In closed loop the spatial lift loses 1.2 percentage points and stays feasible, so the benchmark does not argue for it either. The 47-feature lift is kept.

1 The question, and how it is decided

The lift is the fixed feature vector $\psi(\xi)$ the linear Koopman predictor works on; n_z counts the features. Production uses boundary and consumer measurements only ($n_z = 47$). The question: does adding in-network measurements (junction temperatures, in-pipe temperatures, ring flows), as in Max Sibeijn's paper, predict or control better?

A dictionary is picked on open-loop multi-step prediction error, at the horizons the controller uses ($h \leq 12$) and on the closed-loop distribution it visits (following Korda and Mezić, 2018). The closed loop is only a pass/fail gate here: on one stressed day every leg is pressed against the same capacity ceiling, so comfort is a blunt instrument for separating dictionaries (Section 5).

2 What competed, and the winning lift

Winning lift (production, $n_z = 47$), boundary and consumer measurements only:

block	#	features
thermal states	12	T_{F0}, $T^{(i,s)}$ (5), $T^{(i,r)}$ (5), $T^{(0,r)}$
hydraulic	6	$q^{(i)}$ (5), Q_{net}
delivered-heat coord.	5	$\theta^{(i)} = (T^{(i,s)} - T^{(i,r)})\, q^{(i)}$
constant	1	1
source-delay lags	2	T_{F0} at $t-1$ and $t-2$
delayed bilinears	10	$T_{F0}(t-d)\, q^{(i)}$, 5 consumers $\times$ 2 lags
exergy bilinears	11	$q^{(i)}(T^{(i,s/r)} - T_{\text{amb}})$, $Q_{\text{net}}(T^{(0,r)} - T_{\text{amb}})$
total	47	

$\theta^{(i)}$ makes $c^{(i)} = c_p \theta^{(i)}$ a linear read-out, which keeps the QP convex. The lags and delayed bilinears carry the source transport-delay memory.

Spatial challengers add in-network measurements (the construction of Max Sibeijn's paper): temperatures at the internal junctions where the pipes meet, ring-pipe flows, and flow-weighted node temperatures $q_e\, T_{\mathrm{from}(e)}$. In-pipe temperature is not measured on this benchmark, so the junction temperatures stand in; the literal in-pipe version is in Section 4.

The main spatial challenger (*full*, $n_z = 78$) adds ring temperatures, ring flows and masked products on top of the production lift; standalone and return-side supersets were screened too and rank the same. Each is fitted like the 47: direct multi-step ridge regression, one per consumer and horizon h. The comparison is re-run standardized (unit-variance features) and with the iterated one-step estimator of Max Sibeijn's paper.

3 Result: the 47 wins where it counts (open loop)

Three regimes: held-out PRBS days (broadband, never fitted), a held-out operational day (smooth, at a phase off the training grid), and on-policy (along the stressed benchmark trajectory, `mdot_scale` $= 0.35$). Metric: suite RMSE, delivered-heat error in watts over the five consumers. Each cell is $\mathrm{RMSE}_{\mathrm{spatial}} - \mathrm{RMSE}_{47}$; negative means the spatial lift predicts better.

The on-policy leg is run in both directions. A lift can look good on a trajectory that its own controller produced, so I score both lifts on the closed loop driven by the 47 and on the closed loop driven by the spatial lift. Neither dictionary gets to choose its own operating point. The brackets below are the spread across the trajectories in that regime, not a confidence interval. `eval_dictionary_gap.m` computes every cell from the shipped fits and writes `results/dict_gap.mat`; the figure is drawn from that file.

regime	h	spatial − 47 [W]
PRBS (2 days)	1	-55 $[-65, -45]$ (spatial better)
PRBS (2 days)	12	$+48$ $[+42, +53]$
operational day (held out)	1–12	ties (-15 to $+1$)
on-policy (both trajectories)	8	$+261$ $[+243, +279]$
on-policy (both trajectories)	12	$+222$ $[+216, +228]$

For scale, the 47 lift's own suite RMSE on the on-policy cells is about 740 W at $h = 8$ and 880 W at $h = 12$, so the gap at the deployed horizons is roughly a quarter of the error being made.

Verdict. The spatial lift is more accurate only one step ahead under PRBS. At the deployed horizons, on the trajectory the controller actually visits, the 47 is more accurate, by about +260 W at $h = 8$ and +220 W at $h = 12$, and it is more accurate whichever of the two controllers generated the trajectory. That is the result the choice rests on: the identification regime and the deployment regime disagree, and it is the deployment regime that decides. The standardized fit and the iterated estimator agree.

One honest wrinkle ($h = 12$). The suite number hides a split. At $h = 8$ the 47 is better for every consumer (per-consumer gap +34 to +522 W). At $h = 12$ it is not: the spatial lift wins C1 (−64 W), C3 (−24 W) and C5 (−217 W, markedly), while the 47 wins the two large consumers C2 (+496 W) and C4 (+271 W) that dominate the watt-weighted suite. So the $h = 12$ suite win is carried by the big consumers rather than being uniform, and C5 at $h = 12$ is a cell the spatial lift genuinely takes. It does not change the choice, because C2 and C4 set the comfort outcome and $h = 8$ is clean, but it is the cell to point at if someone asks where the spatial lift would help.

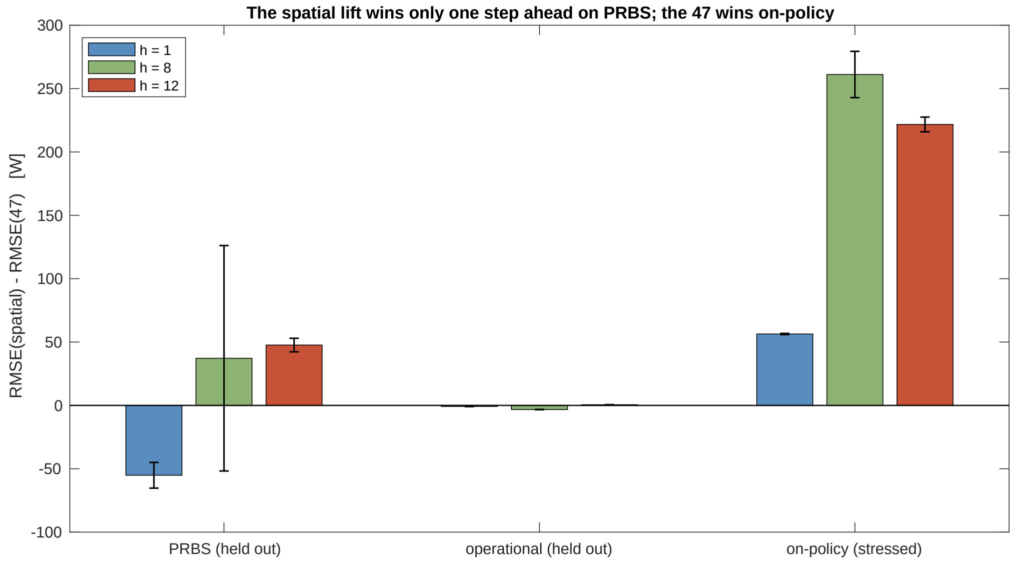

Figure 1: Delivered-heat RMSE of the spatial lift minus the 47 production lift, per regime and horizon. Below zero the spatial lift predicts better, above zero the 47 does. The spatial lift wins only the one-step PRBS fit; at the deployed multi-step horizons and on-policy the 47 predicts better. Whiskers span the trajectories inside each regime.

4 Where the spatial lift wins, conceded

The four variants that ship (`none`, `full`, `nolag`, `return`) are the ones `spatial_library.m` builds and the ones Section 3 scores. This section reports the other things I tried while deciding, and every one of them is a place the spatial construction is better than the 47. They needed lifts and re-simulations that are not part of this package, so the numbers below come from those exploratory runs and cannot be reproduced from what is shipped. I keep them because they are the honest limits of the claim, not because they support it.

Literal in-pipe channels. Re-simulating to capture the true in-pipe temperatures and building the prior construction literally (vs the 47, in W):

	prior, 18 cells	prior, 64 cells	47 + in-pipe
PRBS h=1	+949	+928	−131
on-policy h=8	+3349	+3334	+85
on-policy h=12	+3356	+3370	+134

The literal in-pipe states are more informative one step ahead (−131 vs −59 W) but still significantly worse on-policy. Standalone (without the delivered-heat coordinate and delay lags) it is far worse.

Temperature metric. On the temperature state, the prediction target of Max Sibeijn's paper, the spatial lifts win (0.64–0.66 vs the 47's 0.71–0.73 °C on-policy). The 47 wins on delivered heat, which the controller constrains. Different targets, different winners.

Targeted subset. Adding only the flow-weighted products $q_e T_{\mathrm{from}(e)}$ on the two far branches feeding C3 and C5 (about twelve features) lowers the on-policy suite error by about 4% at $h = 12$ and 1.5% at $h = 8$, on both trajectories. It still does not change the choice: the gain is carried by the large C2 and C4, while the worst-served C5 that sets comfort is significantly worse (+31 W). A lift that helps the suite average but hurts the binding consumer cannot improve worst-consumer comfort.

Mechanism. On the smooth trajectory the boundary measurements plus known future inputs already reconstruct the arriving heat: in a one-dimensional transport network the interior is set by boundary data propagated along the flow once transients wash out (Bastin and Coron, 2016). Each extra feature adds one fitted coefficient, hence estimation variance (Ljung, 1999). One step ahead the interior signal beats that variance; at the deployed horizons the signal has washed downstream while the variance stays, so the net turns negative. The PRBS-learned coefficients also do not transfer to the narrowband regulated distribution (Gevers, 2005). So one-step accuracy need not imply on-policy accuracy, exactly the measured pattern.

5 Closed loop confirms it cannot decide

On the stressed benchmark day (`mdot_scale` = 0.35), both Koopman controllers re-run with the full spatial lift under the same tune, horizon, boxes and floor:

	worst-consumer met [%]	unmet [kWh]
KPC, production lift	97.56	1.03
KPC, spatial lift	96.34	2.18
iterated baseline, either lift	97.74	0.95

With the production tune the spatial controller loses 1.2 percentage points, and it is feasible on every QP step of this stressed day. So the closed loop does not argue for the spatial lift either. But it is a blunt instrument for this question: one stressed day, one tune, and every leg pressed against the same capacity ceiling, which is why a 1.2 pp difference is not by itself a dictionary verdict. The open-loop and on-policy evidence above is what the choice rests on, and it points the same way.

6 How to reproduce

From the repository root:

```
startup
eval_dictionary_gap      % scores both lifts on all three regimes (~5 min)
plot_dictionary_gap      % redraws figure 1 from results/dict_gap.mat
```

`eval_dictionary_gap.m` reads the shipped fits (`vseq_fits_spatial_none.mat` for the 47 and `vseq_fits_spatial_full.mat` for the spatial lift), rebuilds the held-out operational day and the two on-policy closed loops, and writes every number in Section 3. The fits themselves are produced by `koopman/fit_vseq_spatial.m`, which is a longer run.

7 Conclusion

The 47-feature lift is kept. On the trajectory the controller visits, at the horizons it uses, the broad spatial lift predicts delivered heat worse by about +220 to +260 W, and it does so whichever controller generated the trajectory. In closed loop it loses 1.2 percentage points and stays feasible, so the benchmark does not rescue it. The compact lift also wins on parsimony.

Conceded honestly: by the one-step residual and on the temperature metric the spatial lift is better, and it is more interpretable by design. At $h = 12$ it also beats the 47 on three of the five consumers, C5 markedly. By the multi-step control-relevant criterion, on the consumers that set comfort, the 47 wins, and that criterion decides deployment. Two caveats bound the study: the on-policy regime is one deterministic stressed day, and no sensor noise is put on the extra in-network channels, which can only flatter the spatial variants, so this is an upper bound on their value.